

Towards Usable, Privacy Respecting Long-Term Memory for LLM-based Conversational Agents

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Abstract

In recent years, Large Language Models (LLMs) have evolved from stateless text generators into conversational agents (CAs) that use long-term memory to facilitate personalised and compelling interactions. While these emerging capabilities improve user experience, they also introduce ethical and privacy concerns. Although prior work has examined LLM privacy from a technical perspective, few studies have designed features or interaction techniques for supporting users with privacy management. The aim of my thesis is to address this gap by co-designing usable, privacy-respecting memory mechanisms for LLM-based CAs. I present a mixed-methods survey of LLM-based CA users' privacy attitudes and strategies, and an ongoing design taxonomy of conversational privacy controls. I then present a plan for two further studies focused on co-designing and prototyping interface controls with users. I aim for my thesis to contribute tangible mechanisms for enhancing human autonomy in memory-enabled AI interactions.

CCS Concepts

• **Security and privacy** → **Usability in security and privacy**; • **Human-centered computing** → **Empirical studies in interaction design**; • **Computing methodologies** → **Artificial intelligence**.

Keywords

Large Language Models (LLMs), Generative AI, Co-design, Usable Privacy, Memory

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1 Introduction

Generative AI has had wide-reaching impacts across industry, academia, public services, and everyday online life. AI agents such as ChatGPT have evolved beyond impersonal productivity tools to function as assistants and companions [8], driving increasingly intimate user

interactions across work [13], health and fitness [12], and mental well-being [14]. Through these naturalistic interactions, users disclose sensitive information that extends beyond traditional personal data categories to include textured thoughts, feelings, and personal plans [27]. The accumulation of this intimate data, coupled with emerging capabilities in long-term agentic memory, creates significant risks for user privacy and agency.

To clarify how privacy risks from LLM memory emerge, I distinguish between two forms of memory: (i) parametric memory, which comprises information embedded in a model's parameters during pre-training and fine-tuning [28]; and (ii) non-parametric memory, which refers to information stored outside an LLM's weights, such as the contents of previous conversations. Non-parametric information can be dynamically accessed to personalise responses in real-time using techniques such as retrieval augmented generation (RAG) [23]. Prior research on privacy risks in LLM memory has primarily focused on parametric memory and training LLMs on user conversations, and the informed consent and regulatory compliance challenges they pose [8, 26], as well as technical risks such as models leaking training data verbatim [4, 5, 17]. In parallel, HCI studies have revealed a landscape of privacy cynicism and disempowerment among users of ChatGPT. Many users misunderstand the LLM training process, lack awareness of the risks associated with parametric memory, and do not use available mechanisms for opting out of model training [2, 14, 29].

Thus far, research on non-parametric memory has largely focused on proposing architectures for efficient storage, retrieval, and integration of past interactions from a software development, rather than user-focused, perspective (e.g., [7, 20, 21]). However, non-parametric memory has tangible UX consequences: over-referencing prior conversations can elicit discomfort and perceptions of 'creepiness' among users [11], while failing to recall information users consider relevant may undermine trust in an agent's competence [6]. Users often have incomplete mental models of how non-parametric memory mechanisms function, and perceive themselves as having little control over what the system remembers and how memories are referenced [27]. Moreover, non-parametric memory entails the ongoing aggregation, linkage, and summarisation of personal data. These practices can generate sensitive inferences about users [25] and constitute privacy harms if not adequately disclosed and justified [22].

As it currently stands, user-centred privacy research addressing LLM-based CAs remains qualitative and exploratory [13, 14], or



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focuses on perceptual constructs such as privacy concerns and information disclosure [24, 31]. Existing artefact-based research has proposed tools for prompt sanitisation which address explicit personally identifiable information (e.g., names and email addresses) [30] but fail to address implicit disclosures and inference risks [25]. Only a handful of usable privacy studies have addressed parametric or non-parametric memory in depth (see [11, 27, 29]), and though valuable design directions have emerged from these studies, they do not directly propose or evaluate concrete design solutions. Further, existing studies focus almost entirely on ChatGPT, and lack the generalisable, abstract lens necessary to innovate more broadly around the interaction opportunities provided by LLMs.

In sum, the literature suggests that while users express concern about privacy, memory, and inference, they are offered few mechanisms for meaningful control over how LLMs retain and reuse their data. In my thesis, I aim to move beyond exploratory, perceptual studies and towards the co-design of usable, privacy-enhancing memory management tools for LLM-powered interfaces. The expected contributions include: (i) a user-centred, empirical roadmap for operationalising privacy-by-design in LLM-based CAs; and (ii) a set of design artefacts, including interface prototypes, interaction techniques, and system designs, that enhance user understanding of, and control over privacy. In the following sections, I summarise the objectives and expected contributions of this thesis (Section 2). I then describe the work done to date, which includes one first-author publication (Section 3) and an in-progress study which constructs a design space for user control over LLM memory (Section 4). Finally, I provide an overview of my future research directions, which include conducting co-design workshops with users and iterating towards high-fidelity prototypes (Section 5). Throughout, I highlight methodological and conceptual challenges that would most benefit from discussion in the dissertation roundtable session.

2 Research Objectives and Contributions

The aims of my thesis are to understand users' privacy practices, mental models, and needs in relation to LLM memory; and to explore the design space of user-facing controls by co-designing and iteratively prototyping privacy-enhancing interventions with users. Specifically, the aims are:

- (1) **Understanding Users:** To examine users' privacy attitudes, knowledge, and practices related to parametric and non-parametric memory in LLM-based CAs by investigating:
 - Users' engagement with existing privacy controls and settings;
 - Users' understanding of, and concerns about, data storage, model training, and the inferential use of personal information;
 - The strategies users employ to manage information privacy across both memory types, as well as associated pain points and unmet design needs.
- (2) **Exploring the Design Space:** To develop a design taxonomy of user-facing memory control mechanisms for LLM-based CAs, encompassing:
 - Chat history retention and deletion;
 - Mechanisms for controlling the use of conversational data in model fine-tuning or pre-training;

- Interaction logic and interface metaphors for controlling non-parametric memory;

- (3) **Co-designing Privacy Interventions:** To co-design, iteratively develop, and evaluate memory control mechanisms that address users' privacy needs.

This thesis is expected to contribute empirical insights into how end users perceive, understand, and manage information privacy in LLM-based CAs. In particular, I focus on parametric and non-parametric memory functions, which present significant privacy risks yet remain poorly understood from a user-centred perspective. It further contributes a design taxonomy that abstracts existing LLM app privacy controls into generalisable design patterns. To achieve this, I will conduct usability inspections of existing interfaces, allowing me to critique existing design norms, and generate evidence-based priorities for design improvements. Finally, this thesis will advance artefact-based research through the co-design of user-facing memory control mechanisms, and translate AI privacy from an abstract principle into concrete interface design.

3 Study 1: Exploring Privacy Behaviours, Awareness, and Concerns Among UK Users of LLM-Based Conversational Agents (Aim 1)

The first study was published as "Exploring Privacy Behaviours, Awareness, and Concerns Among UK Users of LLM-based Conversational Agents" [15]. Prior qualitative work shows that although users express privacy concerns about LLMs, they often hold inaccurate mental models of parametric memory, have limited awareness of associated risks, and rarely engage with available privacy controls [13, 29]. However, this body of work does not examine risks from inference-based disclosures, nor does it explore users' mental models of specific privacy mechanisms like data deletion or opting out of model training. These gaps are consequential: inference can surface sensitive user insights [25], while misunderstandings of privacy controls undermine their effectiveness [27]. Accordingly, we surveyed LLM-based CA users about their privacy behaviours, self-disclosure boundaries, awareness, and concerns.

Methods. A mixed-methods survey was conducted with 211 UK-based users of LLM-based CAs, including ChatGPT, Microsoft Copilot, Google Gemini, and Claude. Participants, recruited via Prolific, reported using one or more of these apps on a monthly, weekly, or daily basis. We collected demographic information and details about app usage habits, followed by a series of closed-ended questions examining participants' privacy behaviours; usage of in-app privacy features; information disclosure boundaries, and concerns about specific privacy scenarios. These scenarios included LLMs being trained on user conversations, data being leaked from model memory, and sensitive inferences being generated from chats. To probe mental models of privacy features, we presented participants with a mock-up interface resembling the privacy settings and chat deletion interfaces of widely used apps such as ChatGPT. We elicited mental models through closed-ended questions about the expected outcomes of using each feature. To contextualise these findings, participants also answered open-ended questions about day-to-day challenges they encountered when managing app privacy, and potential design or policy interventions. We analysed

close-ended responses using descriptive statistics and logistic regression to examine the effects of demographics and usage patterns on privacy attitudes and behaviours. Open-ended responses were analysed through multiple collaborative rounds of open coding, code refinement, and inter-rater reliability checks.

Findings. The study revealed a complex interplay between privacy knowledge, behaviours, and concerns surrounding data retention and memory. Participants most commonly attempted to self-manage privacy by deleting past conversations and removing personally identifiable or sensitive information from their prompts. Notably, fewer than one in ten participants had opted out of having their data used for model training (i.e., parametric memory). Most participants relied on non-technical mental models of LLM training, incorrectly believing that data could be reliably removed from parametric memory by deleting their accounts. Further, many participants held flawed mental models of chat deletion: the majority assumed that chat data would be permanently erased from company databases immediately or within a few days, conflicting with the standard 30-day retention periods of apps such as ChatGPT [18]. Our qualitative analysis revealed practical barriers to prompt sanitisation: participants felt that removing details made the LLM's responses less useful, and expressed anxiety about sensitive inferences being drawn from information they had not considered private enough to sanitise in isolation.

Privacy concerns were intensified by technical uncertainty and limited transparency into how LLMs operate. While participants articulated worries about data storage, retention, and reuse, they were often uncertain about the nature and likelihood of these risks, and how to control app privacy practices. Participants advocated for stronger regulatory oversight; greater transparency in data use; accessible educational resources; and features that better support privacy self-management, such as granular solutions for opting out of training, and meaningful control over inferences or profiles derived from their interactions. Together, these findings reinforce prior work highlighting the limited usability and scope of existing privacy features in LLM-based CAs [29]. They underscore the need for HCI research to engage more deeply with the privacy design space of LLM-based CAs in order to address growing user disempowerment.

4 Study 2: Evaluating the Design Space of Privacy and Memory Controls for LLM-based CAs (Aim 2)

At the time of Study 1, user-facing support for memory was limited to deleting chat history or opting out of parametric memory (model training), and visible control over non-parametric (retrieval-based) memory in LLM-based CAs was still nascent. Since then, apps such as Gemini, ChatGPT, and Claude have introduced varying degrees of control over the utilisation of prior conversations in an ongoing interaction [3, 9, 19]. ChatGPT for instance, provides the ability to opt-out of past chat references, and to control the automatic extraction and storage of discrete pieces of information for prioritised recall (Saved Memories) [19]. Although a few recent studies have qualitatively examined user perceptions and privacy challenges related to non-parametric memory [11, 27], HCI research

has yet to critically interrogate this emerging design space. The aim of Study 2 is to develop a taxonomy of mechanisms for user control over LLM memory and privacy, drawing from technical literature review and a rigorous usability inspection of existing interfaces.

Methods. The methodology for Study 2 builds on prior work by Zhang et al. [28] and Sumers et al. [23], which represent early attempts at taxonomising long-term memory architectures for LLMs. This prior work introduces high-level distinctions between parametric and non-parametric memory and formalises core memory operations such as creation, retrieval, management, and deletion. Using this taxonomy as an initial analytical framework, I aim to establish more granular design choices and articulate their corresponding UX and privacy implications. To do so, I am conducting a literature review combined with usability inspections of existing mechanisms for managing memory in LLM-based CAs. The literature review covers academic and industrial case studies of memory architectures, as well as user-centred research that provides empirical insights into users' perceptions of and interactions with privacy controls in LLM-based CAs.

The literature review is complemented by usability inspection methods, including cognitive walkthroughs and heuristic evaluation. Usability inspection methods are effective for critiquing existing design norms, and identifying whether users can locate, understand, and accurately anticipate the effects of app features [10]. Such approaches are well suited to evaluating privacy controls and have previously been applied in domains such as secure messaging [1] and intimate health tracking [16]. In this way, I will enrich theoretical taxonomies with case-studies from existing interfaces, and offer best practices for mapping abstract dimensions of LLM memory to interface control mechanisms.

Progress to Date. Study 2 is currently in the literature review phase. An exploratory search has been conducted to develop a preliminary analytical framework that distinguishes between different types of memory in LLM-based systems. This framework also identifies a set of design questions that guide decisions about: (i) how user information is stored and retrieved; (ii) which memory controls are exposed to users; and (iii) the conditions under which an LLM may perform memory operations on a user's behalf.

This initial review has clarified key feasibility challenges related to user control over parametric memory. In particular, data encoded within model weights is difficult to interpret [26], in contrast to non-parametric memory systems, which rely on externally stored textual representations. As a result, non-parametric memory architectures are more amenable to fine-grained, user-facing control, whereas privacy control over parametric memory is largely limited to coarse mechanisms such as opting out of training altogether [28]. Once I have completed a review of academic literature, I will conduct a usability inspection of privacy choice interfaces in existing apps.

Feedback Opportunity. The dissertation research roundtable session offers a valuable opportunity to receive targeted feedback on data analysis, and strategies for synthesising insights from multiple data sources into a coherent taxonomy. At present, the review draws on technical, user-centred, and privacy-focused literature, and will be further enriched with findings from usability inspections. I would value guidance on how to triangulate these sources;

for example, by linking specific technical design decisions (such as data storage, retrieval, and retention strategies) to observed or anticipated user outcomes, including trust, frustration, and privacy concerns [27]. Clearly articulating these dimensions would maximise the value of my research for designers and AI practitioners.

5 Next Steps: Co-designing Control Mechanisms for LLM Memory (Aim 3)

Following Study 2, I will focus on enriching the memory taxonomy with empirical user insights, and develop prototypes of memory control systems that address key usability and privacy challenges. These challenges include: transparent data deletion; supporting opt-out from model training; and designing intuitive workflows for committing chats to memory, prioritising information for retrieval, and making these processes visible and understandable to users [28]. To inform these designs, co-design workshops will be conducted with users of LLM-based CAs from diverse technical backgrounds. Insights from these workshops will guide the development of low-fidelity prototypes, which will then be refined into higher-fidelity systems implemented in Python. These prototypes will be evaluated through controlled experiments and/or think-aloud studies. Throughout this process, the taxonomy developed in Study 2 will be adjusted to reflect new findings.

Feedback Opportunity. During the session, I hope to discuss details of the co-design workshop methodology. Interactions with LLM memory systems are inherently longitudinal, and meaningful user insights may only emerge after weeks or months of use as information accumulates in the system. While single-session workshops can elicit valuable user insights, a longitudinal co-design approach may better-capture how user experience, privacy implications, and system behaviour evolve over time. Balancing time and feasibility constraints with the value of longitudinal co-design is a timely and consequential methodological challenge that would benefit from discussion during the session.

6 Conclusion

In my thesis, I explore the design space for privacy-respecting and user-controllable memory systems in LLM-based CAs. My research includes a user study eliciting privacy perceptions, concerns, and mental models; conceptual work that develops a design taxonomy for memory controls; and co-design workshops that instantiate this taxonomy through high-fidelity design artefacts. Beyond producing tangible design artefacts, the thesis is expected to contribute a user-centred roadmap for enhancing user control over AI systems and mitigating the privacy risks associated with long-term memory. The implications of my research extend beyond text-based conversational interfaces, and encapsulate emerging agentic interactions and sensitive domains such as health and security, where user control and auditability are paramount.

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